Домашнее задание 9.

$(15, 7) \qquad d_{min} = 5$

1) порожд. матрица в дв. виде

2) декодир. 111001100011000

$$t = \left[\frac{d_{min} - 1}{2}\right] = \left[\frac{5-1}{2}\right] = 2$$

$$15 = 2^4 - 1$$

$$GF(2) \to GF(2^4) \qquad p(x) = 1 + x + x^4$$

$\alpha^{15} = 1$, α - примит. эл.

порожд. многочлен имеет корни

$\alpha^1, \alpha^2, \alpha^3, \alpha^4$

α^0	1	0000
α^1	x	0010
α^2	x^2	0100
α^3	x^3	1000
α^4	...	
....		

$$H = \begin{pmatrix}
0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\
0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 \\
0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\
1 & 0 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 1 & 1 & 1 \\
0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\
0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 0 & 1 \\
0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 \\
1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1
\end{pmatrix}$$

$$H = \begin{pmatrix} 1&0&0&0&1&0&1&1&0&0&0&0&0&0&0 \\ 1&1&0&0&1&1&1&0&1&0&0&0&0&0&0 \\ 1&1&1&0&1&1&0&0&0&1&0&0&0&0&0 \\ 0&1&1&1&0&1&1&0&0&0&1&0&0&0&0 \\ 1&0&1&1&0&0&0&0&0&0&0&1&0&0&0 \\ 0&1&0&1&1&0&0&0&0&0&0&0&1&0&0 \\ 0&0&1&0&1&1&0&0&0&0&0&0&0&1&0 \\ 0&0&0&1&0&1&1&0&0&0&0&0&0&0&1 \end{pmatrix}$$

$$H = [A \mid I] \rightarrow G = [I \mid A^T]$$

$$G = \begin{pmatrix} 1&0&0&0&0&0&0&1&1&1&0&1&0&0&0 \\ 0&1&0&0&0&0&0&0&1&1&1&0&1&0&0 \\ 0&0&1&0&0&0&0&0&0&1&1&1&0&1&0 \\ 0&0&0&1&0&0&0&0&0&0&1&1&1&0&1 \\ 0&0&0&0&1&0&0&1&1&1&0&0&1&1&0 \\ 0&0&0&0&0&1&0&0&1&1&1&0&0&1&1 \\ 0&0&0&0&0&0&1&1&1&0&1&0&0&0&1 \end{pmatrix}$$

— <u>порождающая матрица в двоичном виде.</u>

111 0011 00011 000

$$r(x) = x^{14} + x^{13} + x^{12} + x^{9} + x^{8} + x^{4} + x^{3}$$

$$S_1 = r(\alpha) = \alpha^{12}$$

$$S_2 = r(\alpha^2) = \alpha^{9}$$

$$S_3 = r(\alpha^3) = \alpha^{14}$$

$$S_4 = r(\alpha^4) = \alpha^{3}$$

синдромы $\neq 0$ => есть ошибка

$$x = \alpha^{i} \qquad y = \alpha^{j}$$

$$S_1 = x + y$$

$$S_2 = x^2 + y^2$$

$$S_3 = x^3 + y^3$$

$$z^2 + \sigma_1 z + \sigma_2 = 0$$

$$\sigma_1 = S_1 \quad \Rightarrow \quad \sigma_1 = \alpha^{12}$$

$$\sigma_2 = \frac{S_3 + S_1 S_2}{S_1}$$

$$\sigma_2 = \frac{\alpha^{14} + \alpha^{12} \cdot \alpha^{9}}{\alpha^{12}} = \frac{\alpha^{14} + \alpha^{6}}{\alpha^{12}} = \alpha^{-4} = \alpha^{11}$$

$$z^2 + \alpha^{12} z + \alpha^{11} = 0$$

$$(z+1)(z+\alpha^{11}) = 0 \quad \Rightarrow \quad z_1 = 1, \; z_2 = \alpha^{11}$$

=> ошибки в позициях 0 и 11

$$
\begin{array}{r}
+\,111001100011000 \\
000100000000001 \\
\hline
111101100011001
\end{array}
$$

$$c(x) = x^{14}+x^{13}+x^{12}+x^{11}+x^{9}+x^{8}+x^{4}+x^{3}+1$$

$$c(x) = (x^{6}+x^{3}+1)\cdot\underbrace{(x^{8}+x^{7}+x^{6}+x^{4}+1)}_{g(x)}$$

$$m(x) = x^{6}+x^{3}+1$$

=> ИП = $\underline{\underline{„1001001"}}$